



SIMATS

ENGINEERING



SIMATS

Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CAPSTONE PROJECT PRESENTATION

Team - 4

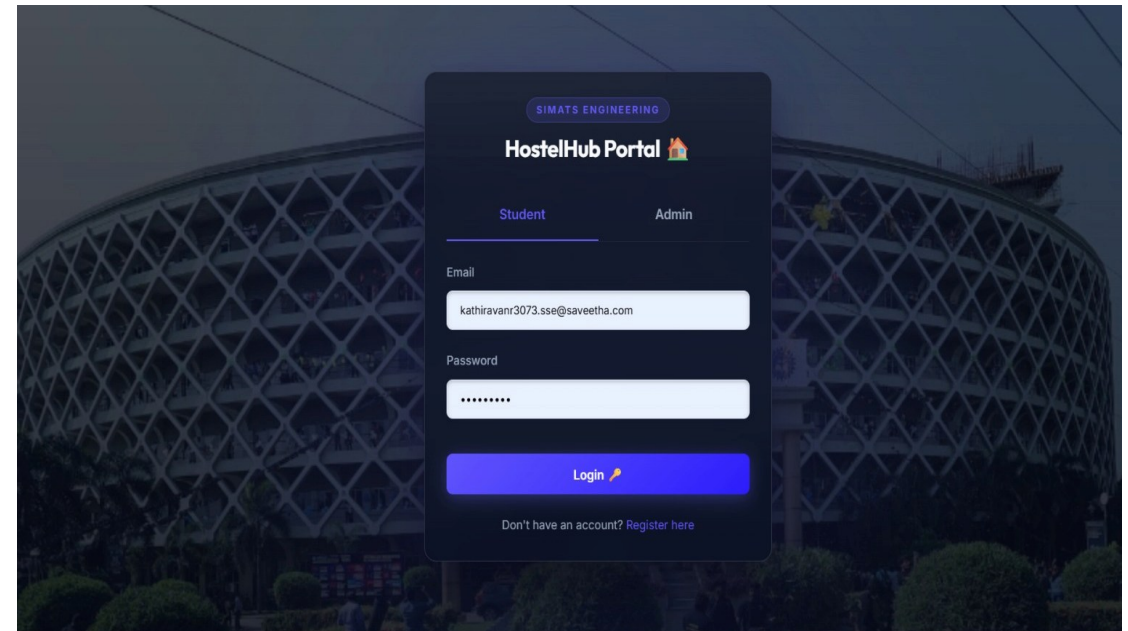
Hostel Room Allotment System

Course Code : **CSA0209**

Course Name : **C - Programming**

Project work Guided by,
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Project work done by,
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ABSTRACT



- Hostel room allotment is usually done manually which consumes time and may result in allocation errors.
- This project automates room allocation based on hostel availability and student information.
- Main Keypoints of this project, Reduces paperwork Faster allocation Easy student management Maintains hostel records digitally



INTRODUCTION



- Project Title: HostelHub – Hostel Room Allotment System
- It is a web-based application designed to automate the hostel room allocation process.
- The project helps students register, apply for rooms, and check their allotment status online.
- It enables administrators to manage student records, room details, and room allotments efficiently.
- The frontend is developed using HTML, CSS, and JavaScript.
- The backend is developed using C programming, and SQLite is used as the database.
- The main goal is to reduce manual work, save time, and improve the accuracy of hostel management.

OBJECTIVES

What This Project Aims to Achieve

- 1 Automate room allocation based on defined rules (availability, preference)
- 2 Maintain accurate, persistent student and room records
- 3 Provide real-time visibility into room occupancy and vacancy
- 4 Reduce manual errors such as duplicate or conflicting allotments
- 5 Give admins an easy way to search, allot, and vacate rooms
- 6 Generate quick reports of hostel occupancy

Problem Statement

1

Time-Consuming

Manually matching students to rooms takes hours and doesn't scale as intake grows.

2

Error-Prone

Duplicate allotments, double bookings, and lost registers are common with paper records.

3

No Real-Time View

Wardens can't instantly see which rooms are vacant, full, or reserved.

4

Hard to Audit

Tracking who stayed where, and for how long, is difficult without digital records.

- Hostel room allocation in many colleges is still performed manually, leading to delays, allocation errors, duplicate assignments, and difficulty in managing room availability.
- An automated Hostel Room Allotment System is needed to efficiently manage student records, allocate rooms, track vacancies, and maintain accurate hostel information while reducing manual effort and improving overall efficiency.

MODULE - 1

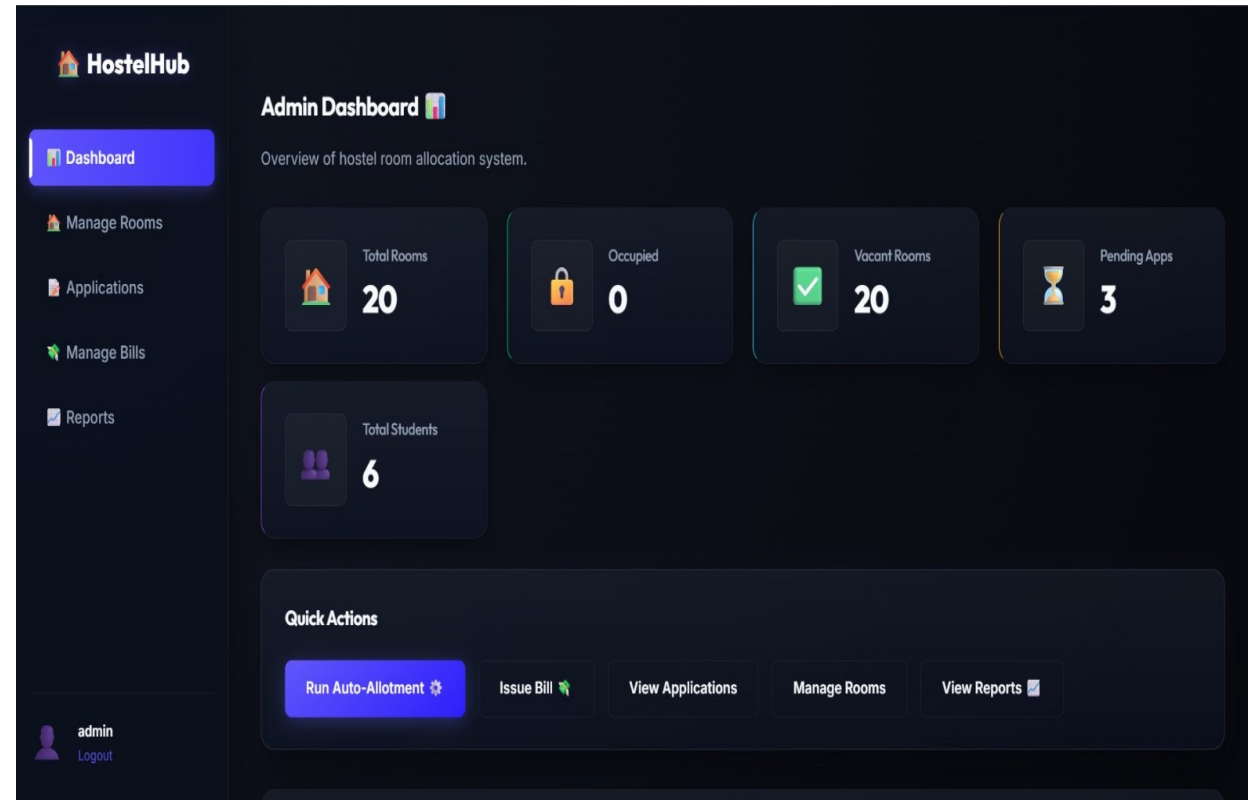
Frontend

Used for designing webpages.

1. HTML
2. CSS
3. JavaScript

Purpose

1. Login page
2. Registration page
3. Dashboard
4. Room selection
5. Reports



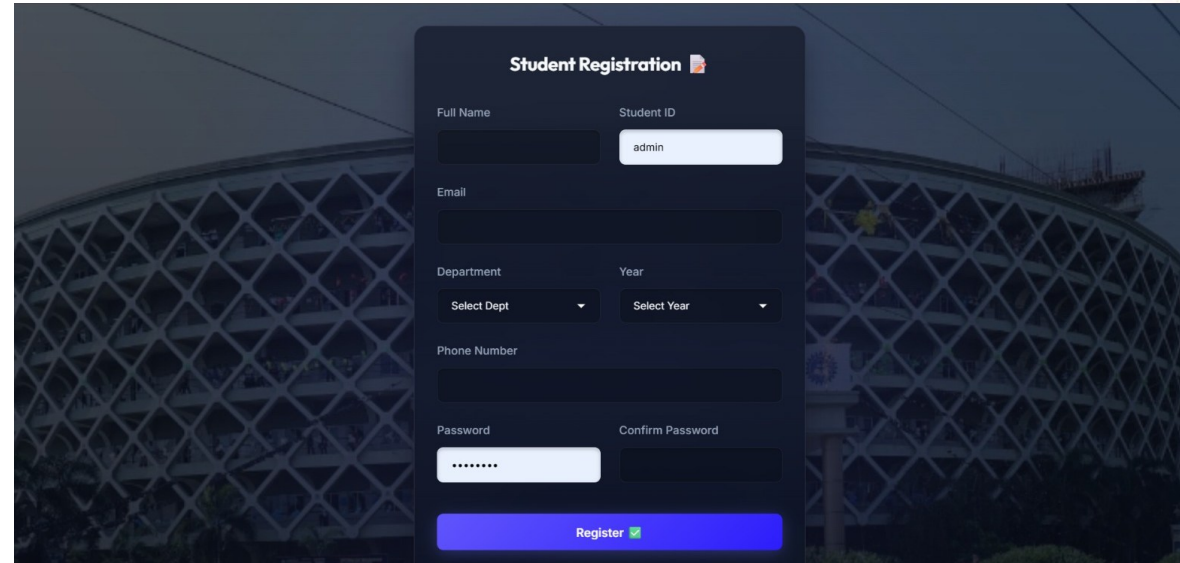
MODULE - 2

Backend

1. Entire backend is written in C Programming.

Purpose

1. Receive requests
2. Validate data
3. Connect database
4. Authenticate users
5. Allocate rooms
6. Send responses



Student Registration

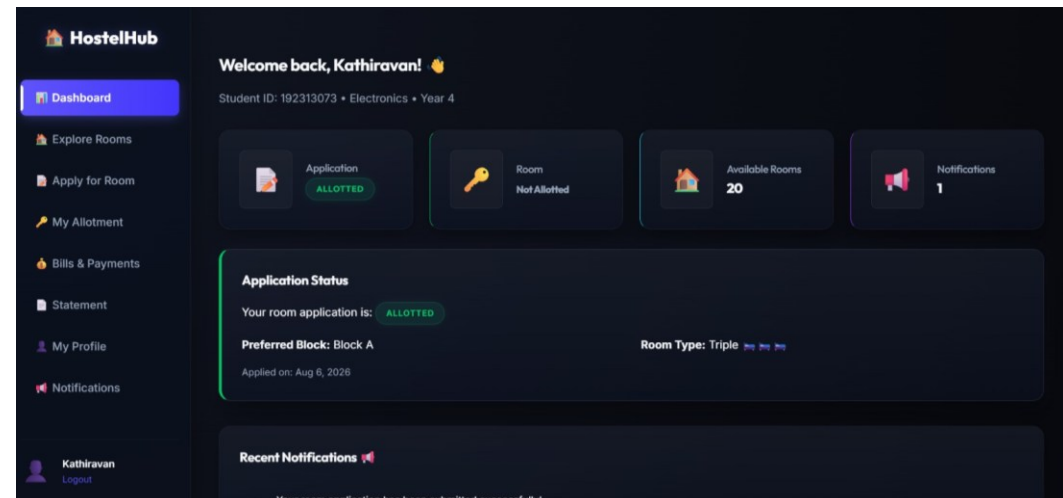
Full Name Student ID

Email

Department Year

Phone Number

Password Confirm Password



HostelHub

Welcome back, Kathiravan! 🏡

Student ID: 192313073 • Electronics • Year 4

Application

Room

Available Rooms 20

Notifications 1

Application Status

Your room application is:

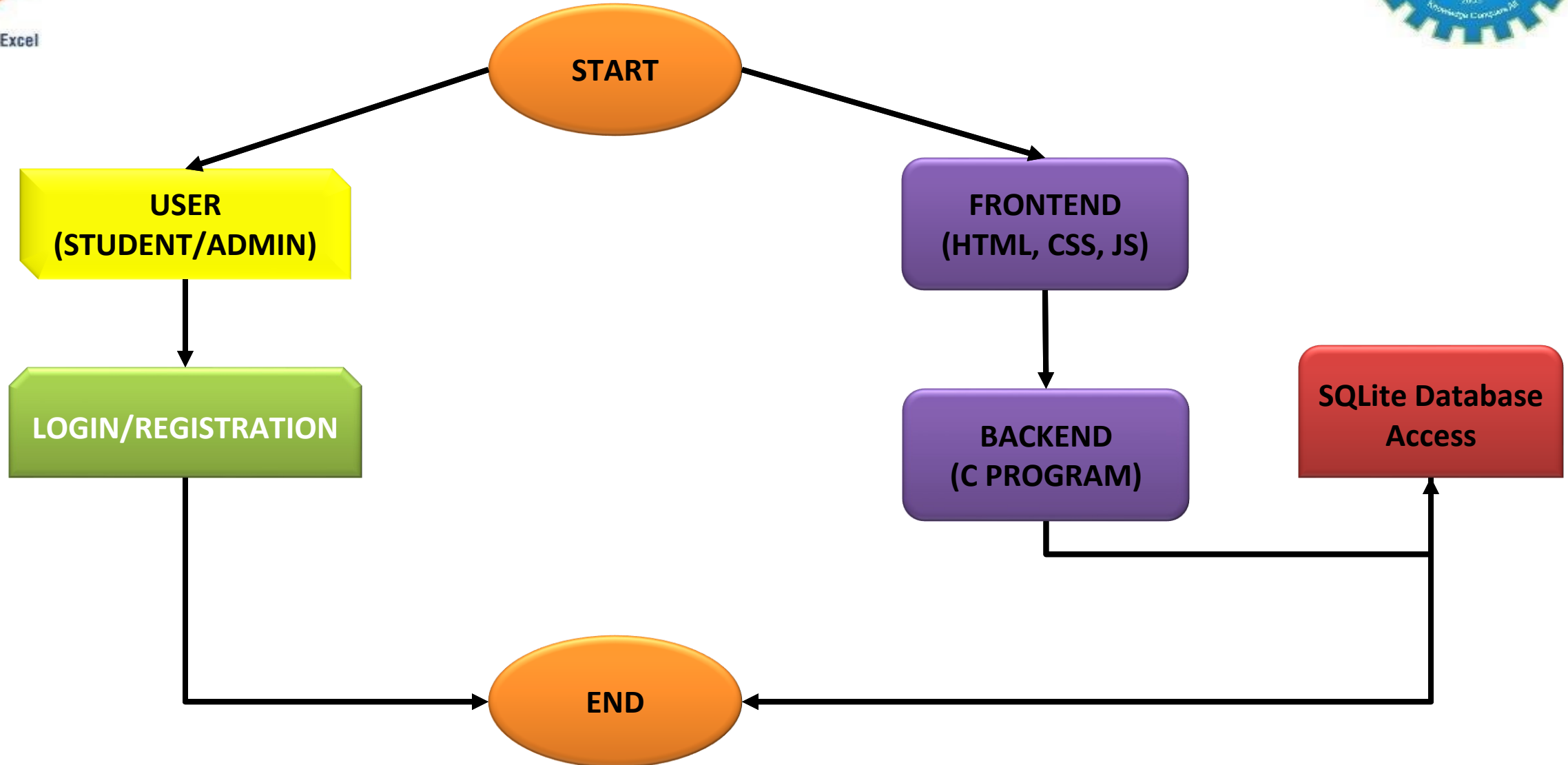
Preferred Block: Block A **Room Type:** Triple

Applied on: Aug 6, 2026

Recent Notifications

Your room application has been submitted successfully!

SYSTEM DESIGN



CORE LOGIC

Data Structures & Allotment Flow

`student.h`

```
struct Student {  
    int id;  
    char name[50];  
    char course[20];  
    int roomAllotted;    // 0 = not allotted  
};
```

```
struct Room {  
    int roomNo;  
    int capacity;  
    int occupied;  
    int studentIds[4];  
};
```

1

Student submits request via frontend form

2

C backend checks room capacity vs occupied count

3

If space available, room is allotted and record updated

4

If full, backend returns next available room or waitlist

5

Room and student files are updated on disk



FEATURES

Core Modules



1

Student Registration

Capture student ID, name, course, and preferences

2

Room Allotment

Assigns rooms based on availability and rules

3

Vacate / De-allot

Frees up a room and updates records instantly

4

Admin Dashboard

Login-protected view for wardens/admin staff

5

Search Records

Look up a student's room or a room's occupant

6

Reports

View occupancy list and vacant room summary



A faster, more reliable way to manage hostel rooms

By combining C's speed and control on the backend with a simple web frontend, this system automates allotment, cuts manual errors, and keeps hostel records accurate and accessible.

CONCLUSION

Welcome Back

Enter details to access hostel operations

Student Portal

Staff Portal

Email Address

e.g. ravi@test.com

Secret Password

.....

 Sign In

New student? [Register Now](#)





Engineer to Excel



Thank You